# Venkata Lakshmi Parimala Pasupuleti

+1-703-261-5410 | venkatalakshmiparimala@gmail.com | linkedin.com/in/venkata-lakshmi-parimala-pasupuleti | github.com/venkatalakshmiparimala | Virginia, USA

## PROFESSIONAL SUMMARY

AI engineer shipping production **LangGraph systems with reproducible eval frameworks** - built **Leasel**, an 8-agent LangGraph pipeline with a 225-query eval harness: 91% Retrieval Precision@1, 81% root-cause Precision@1, 94% Recall@3, pass@k sampling, LLM-as-judge scoring, and regression gating. Also shipped **Magizhs**, an agentic mission-planning system on LangGraph - converted photogrammetry research into a production-grade planner serving marketing, real estate, and commercial survey operations. Deep LangChain/LangGraph/LangSmith expertise: conditional routing, multi-agent state management, HITL escalation, confidence scoring, and guardrails. 8+ years production engineering at Entrata (PropTech, 100K+ MAU); M.S. Data Analytics Engineering, GMU (GPA 3.87, May 2026).

## CORE COMPETENCIES

LangChain / LangGraph / LangSmith | Agentic AI Systems | Eval Frameworks & Monitoring | RAG Pipelines | LLM-as-Judge Scoring | Multi-Agent Orchestration | HITL & Guardrails | Research-to-Production AI

## WORK EXPERIENCE

### Magizhs Tech Solution

May 2025 - Aug 2025

**AI/ML Engineer Intern**

- **Converted quantitative photogrammetry research into a production-grade agentic mission planner** - from algorithm to deployed LangGraph stateful graph (parse → plan → retrieve → validate → conditional routing) serving marketing, real estate, and commercial survey operations.
- RAG layer over FAA Part 107 corpus; every compliance check cited to source regulation; confidence-scored output with ChromaDB upgrade path.
- Engineered **validation guardrails** enforcing altitude limits, battery budgets, and geometric no-fly-zone checks - blocking non-compliant missions before export; improved agent accuracy through iterative prompt engineering.

### George Mason University - College of Engineering

Aug 2025 - May 2026

**Graduate Teaching Assistant (Part-time)**

- Designed and delivered **live technical workshops to 600+ students** across multiple MIS 303 sections - Python programming, AI fundamentals, database design, and SQL; weekly lab instruction and office hours.
- Translated abstract programming concepts into hands-on exercises and one-on-one debugging support for non-technical audiences - the same skill cross-functional AI work demands when communicating technical decisions to diverse stakeholders.

### George Mason University - Research

Aug 2024 - Aug 2025

**Research Associate**

- Achieved $R^2 = 0.977$ across 190+ countries building multi-source prediction models (Python, R) on WHO and UK Road Safety datasets; applied Isolation Forest to surface anomalies manual review missed across 3 research pipelines.
- Improved research comprehension 40% across two cohorts by designing faculty leadership dashboards in Tableau, ggplot2, and Quarto.

### Entrata - PropTech / Real Estate SaaS

Sep 2018 - Aug 2024

**Senior Software Engineer (Remote)**

- **Shipped production code across 20+ full-stack modules** (PHP, Python, React, PostgreSQL, RabbitMQ) across 120+ Agile sprints for 100K+ MAU - 99.5% uptime over 6+ consecutive months.
- **Shipped first production AI pipeline** on 1.6M real financial records (Python, PyTorch, OpenAI API) - converted manual Excel-based research workflow into compliance-gated production delivery; output validation gates, quality controls, and compliance guardrails at launch. Research-to-production is where I live.
- **Optimized 200+ SQL queries** (30% performance gain); led microservices redesign using Docker - 50% scalability improvement, 30% deployment time reduction; AWS S3 migration with 100% PHI compliance, zero security incidents.
- Eliminated 80% of manual operational workload via Python/Bash automation pipelines; mentored 5+ junior engineers across 300+ peer code reviews.

### Entrata India - Pune

Jan 2017 - Sep 2018

**Associate Software Engineer / Trainee**

- Built full-stack property management modules (PHP, Python, PostgreSQL); ETL pipelines for 1.6M HUD/Tax Credit records; cloud migration to AWS S3; redesigned document storage database architecture.

## PROJECTS

### Leasel: Agentic AI Platform - Document Intelligence & Support Investigation 2026

Built an 8-agent **LangGraph** platform with **end-to-end evaluation frameworks and performance monitoring**: **Reproducible eval harness** - 225-query benchmark, 91% Retrieval Precision@1, 81% root-cause Precision@1, 94% Recall@3, pass@k with multi-trial sampling, LLM-as-judge scoring; regression gate blocks degradation before production. **Agent pipeline**: Triage → Retrieval → Configuration → Historical → Reasoning → Validation → HITL Escalation with conditional routing and confidence scoring. **Three surfaces** serving document review, staff investigation, and tenant self-service - built to serve non-technical operators across diverse stakeholder groups. **Open-source** reference implementation on GitHub. Core rule: LLM reasons; deterministic logic or human makes every final regulated call.

Python · LangGraph (8-agent) · LangSmith-compatible eval harness · FastAPI · TF-IDF RAG · OCR · SQLite · LLM-as-Judge · pass@k sampling

### AI on FHIR: NLP-Driven Medical Records Query System 2024

Built a Flask + React NLP platform translating natural language queries (EN/ES) to FHIR APIs using LLMs, BERT, and embeddings - reduced manual clinical retrieval time ~60% and enforced HIPAA-compliant RBAC across structured medical records.

Python · Flask · LLMs · FHIR · React · spaCy · BERT · Embeddings · Docker

## EDUCATION

**M.S. Data Analytics Engineering** · George Mason University, Virginia — May 2026

GPA: 3.87

**B.Tech Computer Science & Engineering** · Jawaharlal Nehru Technological University — 2017

## SKILLS

**Languages:** Python, PHP, JavaScript, TypeScript, SQL, PL/SQL, R, Bash

**AI & Agents:** LangChain, LangGraph, LangSmith, Agentic AI, Multi-Agent Orchestration, Tool Calling, RAG Pipelines, LLM Evaluation, LLM-as-Judge, Eval Frameworks, Performance Monitoring, Inference APIs, Confidence Scoring, HITL, Guardrails, Prompt Engineering, Output Validation, Hallucination Detection, pass@k Sampling, Semantic Search, Embeddings, OpenAI API, Anthropic API, FastAPI, ChromaDB, spaCy, BERT

**Cloud & Infrastructure:** AWS (S3, EC2, RDS, Glue), Azure, Docker, CI/CD (GitHub Actions), Microservices, PostgreSQL, MySQL, Oracle, RabbitMQ, Git, Linux, Agile/Scrum

**ML & Data:** PyTorch, Scikit-learn, XGBoost, Random Forest, Isolation Forest, Feature Engineering, NumPy, Pandas, Tableau, Power BI, ggplot2